

R. Harrison Sweet

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54 Field Brook Rd.
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EXPERIENCE

- Tufts Security and Privacy Lab

SOMERVILLE, MA

Jun 2025 – Present

Research Assistant

 - Interviewed enterprise PSIRT professionals to analyze system-level vulnerabilities and CVSS scoring methodologies for distributed third-party software.
 - Co-authored “Beyond Clinical Risk: An Experimental Study of Cybersecurity Informed Consent and Patient Choice for Connected Medical Devices” in *CHI 2026*, focusing on networked device security.
- Connecticut Judicial Branch

HARTFORD, CT

May 2024 – Aug 2024

IT Intern

 - Developed an external webpage for the DEI department, ensuring compliance with security standards.
 - Analyzed state network governance, gaining direct exposure to enterprise infrastructure risk analysis, hardware provisioning, and incident response.

ENGINEERING PROJECTS

- Containerized Home Network Infrastructure

Jan 2023 – Present

 - Architected a localized server environment on Raspberry Pi hardware, utilizing Docker and Portainer for full-stack service orchestration.
 - Configured secure, encrypted remote access utilizing a WireGuard VPN.
 - Deployed network-wide DNS filtering (AdGuard Home) and integrated telemetry monitoring via Netdata.
- Low-Level Systems & Network Software

Sep 2025 – Jan 2026

 - Engineered a Man-in-the-Middle proxy server natively in C to intercept, route, and manage bi-directional HTTP/HTTPS network traffic.
 - Managed socket-level memory allocation, thread concurrency, and strictly enforced client/server TLS certificate verification protocols.
 - Built a companion Python microservice to parse network payloads through the LLM’s API.
- Encrypted Peer-to-Peer Messaging Protocol

Jan 2025

 - Designed a secure communication script in Python using AES-256-CBC encryption with message authentication verification via SHA-256 HMAC.
 - Managed low-level TCP/IP data transmission, ensuring reliable packet delivery and connection stability across distributed nodes.
 - Integrated cryptographic libraries to enforce end-to-end payload encryption, mitigating packet sniffing and unauthorized data interception.

TECHNICAL SKILLS

Languages & Protocols: C · C++ · Python · Bash · TCP/IP · UDP · HTTP/HTTPS · TLS/SSL
Frameworks & Tools: Linux (Debian) · Docker/Portainer · WireGuard · Tailscale · Git
Core Strengths: Network Architecture · Operating Systems

EDUCATION

- Tufts University

MEDFORD, MA

Sep 2021 – May 2026

 - M.S. in Computer Science (GPA: 3.87), B.A. in Philosophy with a focus on Ethics and Value Theory (GPA: 3.89)
 - Teaching Assistant for *Computation Theory*; mentored 50+ students in formal languages, automata, and abstract problem-solving.

LEADERSHIP & ACTIVITIES

- Varsity Men’s Rowing & Club Ice Hockey

TUFTS UNIVERSITY

Sep 2021 – May 2026

 - Elected captain of both programs, managing logistics for 50+ athletes.
 - Acted as the primary bridge between athletes and university athletic directors for scheduling, facility feedback, and performance milestones.